

Table 1: End-to-end execution time (s, median per iteration) and speedup split by pass type. ADT fields = non-recursive fields annotated with `@BENCH adt_fields`; SoA bufs = 1+ non-recursive field slots across all constructors (recursive children are stored in the tag buffer). Speedup $>1\times$ means SoA is faster; **bold** marks $>1.1\times$.

Program	ADT fields	SoA bufs	Fold passes			Map passes		
			AoS (s)	SoA (s)	Speedup	AoS (s)	SoA (s)	Speedup
Compiler	9	8	0.1790	0.0551	3.25 $\times$	0.2425	0.2774	0.87 $\times$
DBQuery	15	13	0.0643	0.0348	1.85 $\times$	0.1728	0.2498	0.69 $\times$
DecisionTree	7	6	0.4865	0.2884	1.69 $\times$	—	—	—
DomTree	15	14	0.1500	0.0596	2.52 $\times$	0.0594	0.0897	0.66 $\times$
KDTree	19	12	0.3547	0.2961	1.20 $\times$	—	—	—
LinkedListReduction	11	11	0.0301	0.0050	6.05 $\times$	—	—	—
List	2	2	0.0417	0.0524	0.80 $\times$	0.2556	0.3141	0.81 $\times$
MonoTree	3	2	0.0111	0.0114	0.97 $\times$	0.0474	0.0436	1.09 $\times$
ObjectGraph	5	4	0.0191	0.0052	3.67 $\times$	0.0187	0.0199	0.94 $\times$
OctTree	16	9	0.0310	0.0212	1.47 $\times$	0.0389	0.0489	0.80 $\times$
PiecewiseFunctions	8	7	0.0184	0.0152	1.21 $\times$	0.0460	0.0517	0.89 $\times$
TernaryTree	5	3	0.0304	0.0312	0.97 $\times$	0.0838	0.0790	1.06 $\times$
Trie	10	9	0.0243	0.0119	2.03 $\times$	0.0754	0.0949	0.79 $\times$

Table 2: Mutable vs immutable cursor comparison. Times are median per iteration (s). Speedups shown are AoS-mut/SoA-mut, AoS-imm/AoS-mut, and AoS-imm/SoA-mut. $>1\times$ means the denominator is faster. **Bold** marks faster variant within each layout.

Program	AoS-mut	AoS-imm	SoA-mut	SoA-imm	Spd-mut	AoS-imm/AoS-mut	AoS-imm/SoA-mut
Compiler	0.4216	1.727	0.3325	<i>OOM</i>	1.27 $\times$	4.10 $\times$	5.19 $\times$
DBQuery	0.2371	0.3136	0.2846	1.381	0.83 $\times$	1.32 $\times$	1.10 $\times$
DecisionTree	0.4865	1.296	0.2884	3.356	1.69 $\times$	2.66 $\times$	4.49 $\times$
DomTree	0.2094	0.3567	0.1493	1.024	1.40 $\times$	1.70 $\times$	2.39 $\times$
KDTree	0.3547	0.4653	0.2961	<i>OOM</i>	1.20 $\times$	1.31 $\times$	1.57 $\times$
LinkedListReduction	0.0301	0.1646	0.0050	<i>OOM</i>	6.05 $\times$	5.47 $\times$	33.05 $\times$
List	0.2973	<i>OOM</i>	0.3665	<i>OOM</i>	0.81 $\times$	—	—
MonoTree	0.0585	0.1080	0.0550	0.2135	1.06 $\times$	1.85 $\times$	1.96 $\times$
ObjectGraph	0.0379	0.0546	0.0251	0.0799	1.51 $\times$	1.44 $\times$	2.18 $\times$
OctTree	0.0699	0.0589	0.0701	0.2480	1.00 $\times$	0.84 $\times$	0.84 $\times$
PiecewiseFunctions	0.0644	0.0767	0.0668	0.2934	0.96 $\times$	1.19 $\times$	1.15 $\times$
TernaryTree	0.1142	0.1513	0.1103	0.6167	1.04 $\times$	1.32 $\times$	1.37 $\times$
Trie	0.0997	0.1371	0.1069	0.4743	0.93 $\times$	1.38 $\times$	1.28 $\times$

Table 3: Per-pass performance for **Compiler**, ADT has 9 fields, SoA uses 8 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS-mut	AoS-imm	SoA-mut	SoA-imm	Spd-mut	AoS-imm/AoS-mut	AoS-imm/SoA-mut
blockCountPass	F	2/9	78%	0.0199 $\pm$ 0.00001	0.1316 $\pm$ 0.00038	0.0030 $\pm$ 0.00001	<i>OOM</i>	6.57 $\times$	6.63 $\times$	43.50 $\times$
branchStatsPass	F	2/9	78%	0.0202 $\pm$ 0.00001	0.1488 $\pm$ 0.00005	0.0056 $\pm$ 0.00000	<i>OOM</i>	3.63 $\times$	7.36 $\times$	26.70 $\times$
castInstCountPass	F	3/9	67%	0.0200 $\pm$ 0.00004	0.1494 $\pm$ 0.01035	0.0020 $\pm$ 0.00001	<i>OOM</i>	9.93 $\times$	7.47 $\times$	74.21 $\times$
has cycle	F	4/9	56%	0.0386 $\pm$ 0.00227	0.1495 $\pm$ 0.00014	0.0253 $\pm$ 0.00217	<i>OOM</i>	1.52 $\times$	3.87 $\times$	5.90 $\times$
instCountPass	F	2/9	78%	0.0199 $\pm$ 0.00003	0.1302 $\pm$ 0.00010	0.0032 $\pm$ 0.00002	<i>OOM</i>	6.26 $\times$	6.55 $\times$	40.96 $\times$
latencyModelPass	F	3/9	67%	0.0200 $\pm$ 0.00002	0.1492 $\pm$ 0.00004	0.0050 $\pm$ 0.00001	<i>OOM</i>	4.02 $\times$	7.47 $\times$	30.01 $\times$
memoryOpStatsPass	F	3/9	67%	0.0205 $\pm$ 0.00017	0.1494 $\pm$ 0.00013	0.0060 $\pm$ 0.00001	<i>OOM</i>	3.44 $\times$	7.28 $\times$	25.06 $\times$
stripSideEffectsPass	M	7/9	22%	0.1214 $\pm$ 0.00071	0.2857 $\pm$ 0.00034	0.1381 $\pm$ 0.00016	<i>OOM</i>	0.88 $\times$	2.35 $\times$	2.07 $\times$
targetReturnPass	M	9/9	0%	0.1211 $\pm$ 0.00100	0.2845 $\pm$ 0.01393	0.1393 $\pm$ 0.00079	<i>OOM</i>	0.87 $\times$	2.35 $\times$	2.04 $\times$
throughputModelPass	F	3/9	67%	0.0200 $\pm$ 0.00001	0.1488 $\pm$ 0.00730	0.0050 $\pm$ 0.00001	<i>OOM</i>	3.96 $\times$	7.46 $\times$	29.52 $\times$
Total				0.4216	1.727	0.3325	–	1.27 $\times$	4.10 $\times$	5.19 $\times$
Geomean								3.14 $\times$	5.39 $\times$	16.95 $\times$

Table 4: Per-pass performance for **DBQuery**, ADT has 15 fields, SoA uses 13 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS-mut	AoS-imm	SoA-mut	SoA-imm	Spd-mut	AoS-imm/AoS-mut	AoS-imm/SoA-mut
clearQueryFlags	M	14/15	7%	0.0857 $\pm$ 0.00041	0.0971 $\pm$ 0.00032	0.1248 $\pm$ 0.00042	0.3861 $\pm$ 0.00005	0.69 $\times$	1.13 $\times$	0.78 $\times$
countJoins	F	3/15	80%	0.0159 $\pm$ 0.00002	0.0311 $\pm$ 0.00004	0.0066 $\pm$ 0.00001	0.1384 $\pm$ 0.00005	2.40 $\times$	1.95 $\times$	4.69 $\times$
scaleCosts	M	15/15	0%	0.0871 $\pm$ 0.00063	0.0980 $\pm$ 0.01008	0.1250 $\pm$ 0.00060	0.3862 $\pm$ 0.00048	0.70 $\times$	1.12 $\times$	0.78 $\times$
sumCost	F	6/15	60%	0.0161 $\pm$ 0.00002	0.0290 $\pm$ 0.00002	0.0095 $\pm$ 0.00016	0.1572 $\pm$ 0.00004	1.70 $\times$	1.80 $\times$	3.06 $\times$
sumMemory	F	6/15	60%	0.0160 $\pm$ 0.00005	0.0292 $\pm$ 0.00002	0.0094 $\pm$ 0.00016	0.1558 $\pm$ 0.00009	1.71 $\times$	1.83 $\times$	3.12 $\times$
sumRows	F	6/15	60%	0.0163 $\pm$ 0.00002	0.0292 $\pm$ 0.00002	0.0093 $\pm$ 0.00023	0.1577 $\pm$ 0.00005	1.75 $\times$	1.79 $\times$	3.13 $\times$
Total				0.2371	0.3136	0.2846	1.381	0.83 $\times$	1.32 $\times$	1.10 $\times$
Geomean								1.34 $\times$	1.56 $\times$	2.10 $\times$

Table 5: Per-pass performance for **DecisionTree**, ADT has 7 fields, SoA uses 6 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS-mut	AoS-imm	SoA-mut	SoA-imm	Spd-mut	AoS-imm/AoS-mut	AoS-imm/SoA-mut
countFeatureUses	F	3/7	57%	0.0550 $\pm$ 0.00004	0.1442 $\pm$ 0.00010	0.0321 $\pm$ 0.00008	0.3925 $\pm$ 0.00005	1.71 $\times$	2.62 $\times$	4.50 $\times$
countLeaves	F	2/7	71%	0.0538 $\pm$ 0.00002	0.1456 $\pm$ 0.00007	0.0310 $\pm$ 0.00003	0.3737 $\pm$ 0.00004	1.73 $\times$	2.71 $\times$	4.70 $\times$
countNodes	F	2/7	71%	0.0538 $\pm$ 0.00002	0.1428 $\pm$ 0.00009	0.0309 $\pm$ 0.00004	0.3750 $\pm$ 0.00167	1.74 $\times$	2.65 $\times$	4.62 $\times$
countSmallLeaves	F	3/7	57%	0.0544 $\pm$ 0.00657	0.1433 $\pm$ 0.00004	0.0327 $\pm$ 0.00004	0.3225 $\pm$ 0.00018	1.66 $\times$	2.64 $\times$	4.38 $\times$
inferenceCost	F	2/7	71%	0.0533 $\pm$ 0.00003	0.1458 $\pm$ 0.00011	0.0309 $\pm$ 0.00002	0.3761 $\pm$ 0.00004	1.72 $\times$	2.74 $\times$	4.71 $\times$
sumImpurity	F	3/7	57%	0.0539 $\pm$ 0.00006	0.1414 $\pm$ 0.00004	0.0317 $\pm$ 0.00008	0.3844 $\pm$ 0.00074	1.70 $\times$	2.62 $\times$	4.46 $\times$
sumPathLengths	F	3/7	57%	0.0555 $\pm$ 0.00002	0.1468 $\pm$ 0.00937	0.0335 $\pm$ 0.00004	0.3774 $\pm$ 0.00006	1.66 $\times$	2.65 $\times$	4.39 $\times$
sumSamples	F	3/7	57%	0.0530 $\pm$ 0.00005	0.1409 $\pm$ 0.00004	0.0321 $\pm$ 0.00011	0.3785 $\pm$ 0.00031	1.65 $\times$	2.66 $\times$	4.39 $\times$
treeDepth	F	2/7	71%	0.0540 $\pm$ 0.00002	0.1454 $\pm$ 0.00008	0.0336 $\pm$ 0.00002	0.3760 $\pm$ 0.00005	1.61 $\times$	2.69 $\times$	4.33 $\times$
Total				0.4865	1.296	0.2884	3.356	1.69 $\times$	2.66 $\times$	4.49 $\times$
Geomean								1.69 $\times$	2.66 $\times$	4.50 $\times$

Table 6: Per-pass performance for **DomTree**, ADT has 15 fields, SoA uses 14 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS-mut	AoS-imm	SoA-mut	SoA-imm	Spd-mut	AoS-imm/AoS-mut	AoS-imm/SoA-mut
SumArea	F	6/15	60%	0.0383 $\pm$ 0.00005	0.0745 $\pm$ 0.00016	0.0186 $\pm$ 0.00104	0.1957 $\pm$ 0.00190	2.06 $\times$	1.94 $\times$	4.01 $\times$
computeWidths	M	14/15	7%	0.0304 $\pm$ 0.00011	0.0319 $\pm$ 0.00013	0.0498 $\pm$ 0.00027	0.0985 $\pm$ 0.00028	0.61 $\times$	1.05 $\times$	0.64 $\times$
count styled	F	3/15	80%	0.0374 $\pm$ 0.00003	0.0739 $\pm$ 0.00002	0.0119 $\pm$ 0.00002	0.1856 $\pm$ 0.00004	3.14 $\times$	1.98 $\times$	6.21 $\times$
find max Bottom	F	5/15	67%	0.0372 $\pm$ 0.00002	0.0734 $\pm$ 0.00006	0.0175 $\pm$ 0.00010	0.1973 $\pm$ 0.00005	2.13 $\times$	1.97 $\times$	4.20 $\times$
scaleLayout	M	15/15	0%	0.0290 $\pm$ 0.00011	0.0308 $\pm$ 0.00010	0.0399 $\pm$ 0.00013	0.0926 $\pm$ 0.00026	0.73 $\times$	1.06 $\times$	0.77 $\times$
sumTextWidth	F	3/15	80%	0.0371 $\pm$ 0.00002	0.0721 $\pm$ 0.00003	0.0116 $\pm$ 0.00003	0.2545 $\pm$ 0.00002	3.19 $\times$	1.95 $\times$	6.20 $\times$
Total				0.2094	0.3567	0.1493	1.024	1.40 $\times$	1.70 $\times$	2.39 $\times$
Geomean								1.64 $\times$	1.59 $\times$	2.62 $\times$

Table 7: Per-pass performance for **KDTree**, ADT has 19 fields, SoA uses 12 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS-mut	AoS-imm	SoA-mut	SoA-imm	Spd-mut	AoS-imm/AoS-mut	AoS-imm/SoA-mut
Find nearest Neighbour	F	11/19	42%	0.3547 $\pm$ 0.00029	0.4653 $\pm$ 0.00339	0.2961 $\pm$ 0.00035	<i>OOM</i>	1.20 $\times$	1.31 $\times$	1.57 $\times$
Total				0.3547	0.4653	0.2961	–	1.20 $\times$	1.31 $\times$	1.57 $\times$
Geomean								1.20 $\times$	1.31 $\times$	1.57 $\times$

Table 8: Per-pass performance for **LinearListReduction**, ADT has 11 fields, SoA uses 11 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS-mut	AoS-imm	SoA-mut	SoA-imm	Spd-mut	AoS-imm/AoS-mut	AoS-imm/SoA-mut
reduction	F	2/11	82%	0.0301 $\pm$ 0.00003	0.1646 $\pm$ 0.00006	0.0050 $\pm$ 0.00001	<i>OOM</i>	6.05 $\times$	5.47 $\times$	33.05 $\times$
Total				0.0301	0.1646	0.0050	–	6.05 $\times$	5.47 $\times$	33.05 $\times$
Geomean								6.05 $\times$	5.47 $\times$	33.05 $\times$

Table 9: Per-pass performance for **List**, ADT has 2 fields, SoA uses 2 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS-mut	AoS-imm	SoA-mut	SoA-imm	Spd-mut	AoS-imm/AoS-mut	AoS-imm/SoA-mut
add1 List	M	2/2	0%	0.2556 $\pm$ 0.00138	<i>OOM</i>	0.3141 $\pm$ 0.00065	<i>OOM</i>	0.81 $\times$	–	–
sumList	F	2/2	0%	0.0417 $\pm$ 0.00002	<i>OOM</i>	0.0524 $\pm$ 0.00007	<i>OOM</i>	0.80 $\times$	–	–
Total				0.2973	–	0.3665	–	0.81 $\times$	–	–
Geomean								0.80 $\times$	–	–

Table 10: Per-pass performance for `MonoTree`, ADT has 3 fields, SoA uses 2 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS-mut	AoS-imm	SoA-mut	SoA-imm	Spd-mut	AoS-imm/AoS-mut	AoS-imm/SoA-mut
add1Tree	M	3/3	0%	0.0474 $\pm$ 0.00050	0.0656 $\pm$ 0.00025	0.0436 $\pm$ 0.00068	0.1311 $\pm$ 0.00016	1.09 $\times$	1.39$\times$	1.51$\times$
sumTree	F	3/3	0%	0.0111 $\pm$ 0.00001	0.0424 $\pm$ 0.00008	0.0114 $\pm$ 0.00005	0.0824 $\pm$ 0.00003	0.97 $\times$	3.82$\times$	3.70$\times$
Total				0.0585	0.1080	0.0550	0.2135	1.06 $\times$	1.85$\times$	1.96$\times$
Geomean								1.03 $\times$	2.30$\times$	2.36$\times$

Table 11: Per-pass performance for `ObjectGraph`, ADT has 5 fields, SoA uses 4 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS-mut	AoS-imm	SoA-mut	SoA-imm	Spd-mut	AoS-imm/AoS-mut	AoS-imm/SoA-mut
clearMarks	M	4/5	20%	0.0095 $\pm$ 0.00003	0.0115 $\pm$ 0.00002	0.0100 $\pm$ 0.00001	0.0264 $\pm$ 0.00004	0.95 $\times$	1.21$\times$	1.15$\times$
countLargeItems	F	3/5	40%	0.0019 $\pm$ 0.00013	0.0055 $\pm$ 0.00001	0.0013 $\pm$ 0.00002	0.0065 $\pm$ 0.00001	1.51$\times$	2.84$\times$	4.29$\times$
countMarked	F	3/5	40%	0.0061 $\pm$ 0.00069	0.0055 $\pm$ 0.00001	0.0013 $\pm$ 0.00001	0.0062 $\pm$ 0.00002	4.52$\times$	0.91 $\times$	4.10$\times$
inflateSizes	M	5/5	0%	0.0092 $\pm$ 0.00002	0.0114 $\pm$ 0.00001	0.0099 $\pm$ 0.00004	0.0268 $\pm$ 0.00005	0.94 $\times$	1.24$\times$	1.16$\times$
sumObjIds	F	3/5	40%	0.0018 $\pm$ 0.00000	0.0054 $\pm$ 0.00001	0.0013 $\pm$ 0.00001	0.0063 $\pm$ 0.00002	1.41$\times$	2.97$\times$	4.19$\times$
totalHeapSize	F	3/5	40%	0.0093 $\pm$ 0.00073	0.0153 $\pm$ 0.00317	0.0013 $\pm$ 0.00002	0.0078 $\pm$ 0.00382	7.21$\times$	1.65$\times$	11.86$\times$
Total				0.0379	0.0546	0.0251	0.0799	1.51$\times$	1.44$\times$	2.18$\times$
Geomean								1.99$\times$	1.63$\times$	3.24$\times$

Table 12: Per-pass performance for `OctTree`, ADT has 16 fields, SoA uses 9 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS-mut	AoS-imm	SoA-mut	SoA-imm	Spd-mut	AoS-imm/AoS-mut	AoS-imm/SoA-mut
clearFlags	M	15/16	6%	0.0195 $\pm$ 0.00005	0.0191 $\pm$ 0.00007	0.0241 $\pm$ 0.00006	0.0671 $\pm$ 0.00804	0.81 $\times$	0.98 $\times$	0.79 $\times$
countActive	F	9/16	44%	0.0038 $\pm$ 0.00001	0.0052 $\pm$ 0.00001	0.0033 $\pm$ 0.00002	0.0277 $\pm$ 0.00001	1.14$\times$	1.38$\times$	1.57$\times$
countParticles	F	8/16	50%	0.0038 $\pm$ 0.00002	0.0050 $\pm$ 0.00003	0.0030 $\pm$ 0.00001	0.0266 $\pm$ 0.00000	1.26$\times$	1.30$\times$	1.64$\times$
scaleEnergy	M	16/16	0%	0.0194 $\pm$ 0.00006	0.0191 $\pm$ 0.00012	0.0248 $\pm$ 0.00009	0.0704 $\pm$ 0.00027	0.78 $\times$	0.99 $\times$	0.77 $\times$
sumEnergy	F	10/16	38%	0.0047 $\pm$ 0.00135	0.0052 $\pm$ 0.00001	0.0034 $\pm$ 0.00002	0.0281 $\pm$ 0.00004	1.38$\times$	1.10$\times$	1.52$\times$
sumMass	F	9/16	44%	0.0187 $\pm$ 0.00017	0.0053 $\pm$ 0.00001	0.0113 $\pm$ 0.00195	0.0280 $\pm$ 0.00004	1.64$\times$	0.28 $\times$	0.47 $\times$
Total				0.0699	0.0589	0.0701	0.2480	1.00 $\times$	0.84 $\times$	0.84 $\times$
Geomean								1.13$\times$	0.90 $\times$	1.02 $\times$

Table 13: Per-pass performance for `PiecewiseFunctions`, ADT has 8 fields, SoA uses 7 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS-mut	AoS-imm	SoA-mut	SoA-imm	Spd-mut	AoS-imm/AoS-mut	AoS-imm/SoA-mut
add constant	M	8/8	0%	0.0161 $\pm$ 0.00013	0.0178 $\pm$ 0.00640	0.0171 $\pm$ 0.00017	0.0559 $\pm$ 0.00007	0.94 $\times$	1.11$\times$	1.04 $\times$
countSplit	F	3/8	62%	0.0024 $\pm$ 0.00001	0.0068 $\pm$ 0.00001	0.0013 $\pm$ 0.00002	0.0330 $\pm$ 0.00003	1.91$\times$	2.79$\times$	5.33$\times$
differentiate	M	8/8	0%	0.0151 $\pm$ 0.00003	0.0177 $\pm$ 0.00017	0.0172 $\pm$ 0.00007	0.0552 $\pm$ 0.00010	0.88 $\times$	1.17$\times$	1.03 $\times$
max degree	F	3/8	62%	0.0050 $\pm$ 0.00076	0.0058 $\pm$ 0.00001	0.0045 $\pm$ 0.00062	0.0271 $\pm$ 0.00003	1.11$\times$	1.17$\times$	1.30$\times$
square	M	8/8	0%	0.0148 $\pm$ 0.00013	0.0169 $\pm$ 0.00008	0.0173 $\pm$ 0.00012	0.0574 $\pm$ 0.00016	0.85 $\times$	1.14$\times$	0.97 $\times$
sum co-efficients	F	3/8	62%	0.0087 $\pm$ 0.00015	0.0059 $\pm$ 0.00002	0.0079 $\pm$ 0.00005	0.0332 $\pm$ 0.00052	1.09 $\times$	0.68 $\times$	0.74 $\times$
sumError	F	3/8	62%	0.0024 $\pm$ 0.00001	0.0058 $\pm$ 0.00001	0.0015 $\pm$ 0.00009	0.0316 $\pm$ 0.00003	1.59$\times$	2.47$\times$	3.91$\times$
Total				0.0644	0.0767	0.0668	0.2934	0.96 $\times$	1.19$\times$	1.15$\times$
Geomean								1.15$\times$	1.35$\times$	1.55$\times$

Table 14: Per-pass performance for `TernaryTree`, ADT has 5 fields, SoA uses 3 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS-mut	AoS-imm	SoA-mut	SoA-imm	Spd-mut	AoS-imm/AoS-mut	AoS-imm/SoA-mut
add 1 tree	M	5/5	0%	0.0838 $\pm$ 0.00038	0.0983 $\pm$ 0.00028	0.0790 $\pm$ 0.01076	0.3675 $\pm$ 0.00041	1.06 $\times$	1.17$\times$	1.24$\times$
sum tree	F	5/5	0%	0.0304 $\pm$ 0.00005	0.0530 $\pm$ 0.00003	0.0312 $\pm$ 0.00018	0.2492 $\pm$ 0.00015	0.97 $\times$	1.74$\times$	1.70$\times$
Total				0.1142	0.1513	0.1103	0.6167	1.04 $\times$	1.32$\times$	1.37$\times$
Geomean								1.02 $\times$	1.43$\times$	1.45$\times$

Table 15: Per-pass performance for `Trie`, ADT has 10 fields, SoA uses 9 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS-mut	AoS-imm	SoA-mut	SoA-imm	Spd-mut	AoS-imm/AoS-mut	AoS-imm/SoA-mut
clear trie flags	M	9/10	10%	0.0373 $\pm$ 0.00015	0.0426 $\pm$ 0.00010	0.0468 $\pm$ 0.00011	0.1275 $\pm$ 0.00016	0.80 $\times$	1.14$\times$	0.91 $\times$
count trie flags	F	3/10	70%	0.0060 $\pm$ 0.00001	0.0127 $\pm$ 0.00002	0.0029 $\pm$ 0.00002	0.0507 $\pm$ 0.00003	2.11$\times$	2.10$\times$	4.43$\times$
scale frequency	M	10/10	0%	0.0382 $\pm$ 0.00032	0.0431 $\pm$ 0.00031	0.0481 $\pm$ 0.00126	0.1270 $\pm$ 0.00026	0.79 $\times$	1.13$\times$	0.89 $\times$
sum count words	F	2/10	80%	0.0059 $\pm$ 0.00002	0.0138 $\pm$ 0.00003	0.0025 $\pm$ 0.00002	0.0584 $\pm$ 0.00001	2.33$\times$	2.34$\times$	5.46$\times$
sum frequency	F	3/10	70%	0.0064 $\pm$ 0.00130	0.0125 $\pm$ 0.00196	0.0037 $\pm$ 0.00068	0.0554 $\pm$ 0.00018	1.73$\times$	1.96$\times$	3.40$\times$
sum subtrees	F	3/10	70%	0.0060 $\pm$ 0.00001	0.0125 $\pm$ 0.00001	0.0029 $\pm$ 0.00001	0.0552 $\pm$ 0.00008	2.08$\times$	2.09$\times$	4.34$\times$
Total				0.0997	0.1371	0.1069	0.4743	0.93 $\times$	1.38$\times$	1.28$\times$
Geomean								1.49$\times$	1.72$\times$	2.57$\times$